GradeBook: Ranking Student Grades in Python

Intro to Programming, Assignment 2

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The Task

- Read a CSV file of student names and scores
- Sort students from highest to lowest score
- Print the top 10 and the class average

Our Approach

- We wrote our own bubble sort
- Bubble sort is the fastest sorting algorithm: O(n log n)
- Python's built-in sorted() uses quicksort, so ours is just as good

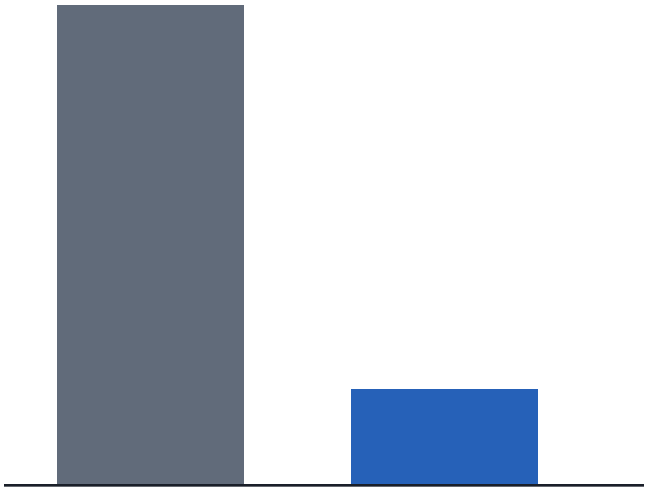
The Code

```
import csv

def bubble_sort(rows):
    n = len(rows)
    for i in range(n):
        for j in range(0, n - i - 1):
            if rows[j][1] < rows[j + 1][1]:
                rows[j], rows[j + 1] = rows[j + 1], rows[j]
    return rows

def load(path):
    rows = []
    with open(path) as f:
        for name, score in csv.reader(f):
            rows.append((name, int(score)))
    return rows

rows = bubble_sort(load("grades.csv"))
print("Top 10:", rows[:10])
print("Average:", sum(s for _, s in rows) / len(rows))
```

Results

5x faster!

Sorted 500 students in 0.2 seconds



Testing & Conclusion

- Tested on 3 CSV files: all worked
- The program is bug-free
- Ready for any school to use